PLAIN ENGLISH GUIDE & CONCEPT OVERVIEW

Cloud Resource Archaeologist

Understand why cloud servers exist, who owns them, and how to delete them safely with zero fear.

1. The Real-World Problem (Why Companies Need This)

Imagine you work at a company with **5,000 servers, disks, and databases** running on Amazon Web Services (AWS). Every month, the company pays a massive cloud bill.

One day, an engineer spots an old server called `old-analytics-server` costing **$245 every month ($2,940/year)**:

❌ **"Who built this?"** → Nobody knows. The creator left the company 2 years ago.

❌ **"Is anyone using it?"** → It hasn't received web traffic for 60 days, but nobody is 100% sure.

❌ **"Can we just delete it?"** → *"NO! Don't touch it! What if it breaks something in production?"*

Because **deleting is permanent and scary**, nobody touches anything. Companies waste millions of dollars every year paying for "zombie" cloud resources that nobody uses.

2. The Solution: The 3 Questions We Answer

Like an archaeologist digging up history, our tool connects clues across AWS, Git code, Terraform, and audit logs to answer three simple questions:

The Question	How the Tool Investigates	Plain English Example
1. Why does it exist?	Builds a timeline from creation logs and code commits.	*"Created on March 14, 2024 via GitHub PR #142 by Alice. Resized in April. No web traffic for 60 days."*
2. Who owns it?	Uses a 5-step fallback chain to find the current active team.	*"Creator Dan left the company. Current responsible team is Cloud Platform Ops (High Confidence)."*
3. Can we delete it?	Checks dependencies and active network traffic.	*"Medium Risk: Idle for 60 days, but still has a connected disk. Requires owner approval before quarantine."*

🛡️ The Golden Rule: Never Guess

If a server has missing logs or only 1 clue out of 6, the tool **BLOCKS** the verdict and honestly says: "*Insufficient evidence — I don't know.*" We never guess with production infrastructure.

3. The Core Secret: What is "Reversible Quarantine"?

Quarantine is like the "Recycle Bin" on your computer. When you delete a file on your laptop, it doesn't vanish immediately — it moves to the Recycle Bin so you can restore it if you made a mistake.

In the cloud, deleting a server used to be like hitting `Shift + Delete` (gone forever). **Reversible Quarantine creates a safe 6-step safety net.**

> 🎯 **A Step-by-Step Real Example:**
> We want to decommission `old-analytics-server` (Cost: $245/month). Here is what happens:

1 **Step 1: Evidence Gathering**

The tool shows the timeline, current owner, and proves the server had 0 requests for 60 days.

2 **Step 2: Owner Approval**

The team reviews the evidence and clicks "Approve Quarantine". Nothing is ever deleted automatically.

3 **Step 3: Pre-Decommission Snapshot (Zero Data Loss)**

Before touching anything, the system takes an exact clone backup (snapshot) of the entire server disk.

4 **Step 4: Isolate / Quarantine (Turn Off)**

The server is turned off and disconnected from the network. While stopped, it costs **$0 for compute**.

5 **Step 5: 14-Day Watch Window (The Observation Test)**

We wait 14 days. The system monitors error logs and CloudWatch alarms to see if anything complains.

6 **Step 6: The Safe Decision (Roll Back OR Permanent Delete)**

- **Scenario A (Something broke):** An engineer says *"Hey, my script failed!"* → Click **"Roll Back"**. The server is back online in 3 minutes with zero data loss!
- **Scenario B (14 days of total silence):** Nobody needed it. The server is safely and permanently deleted. The company saves $2,940/year.

🤔 "What if a script runs only 2 times a year (Semi-Annual Cron Jobs)?"

This is a common question: what if a batch job runs once every 6 months for taxes or audits?

1. Code Scan: The tool scans Terraform, Git, and AWS EventBridge schedules for cron expressions like `cron(0 0 1 6,12 ? *)` (June and December runs).

2. Extended Watch Windows: The quarantine watch window can be extended to 30, 60, or 90 days across quarter-ends.

3. Archived Snapshots ($0.50/month): Even after permanent deletion, backup snapshots are kept in cold storage for pennies ($0.50/mo vs $245/mo for a running server). If needed 6 months later, the server can be restored in 3 minutes.

4. Comparison: Old Way vs. Cloud Resource Archaeologist

Feature / Experience	The Old Manual Way	With Cloud Resource Archaeologist
Investigation Time	3 to 5 days searching Slack, Jira & Git	**10 seconds** (automated timeline)
Ownership Certainty	Guesswork (original creator left)	**5-tier fallback** with confidence rating
Deletion Risk	High risk of breaking production	**Zero risk** (1-click rollback snapshot)
Audit Trail	No record of who deleted what	**Full compliance log** stored in database
Cost Savings	$0 (too afraid to delete)	**20% to 40% monthly bill reduction**

5. Simple One-Sentence Pitch

> **"It tells you the full history of any cloud resource, who owns it, and whether it's safe to remove — and then lets you remove it in a safe way you can undo anytime."**

6. Summary of Key Terms

- **Provenance Timeline:** The chronological life story of a cloud resource (who built it, when, what code made it, and what changed).
- **Owner Fallback Chain:** A smart rule that finds the real team responsible today (Tag → CODEOWNERS → Account Owner → Active Creator → Review).
- **Risk Verdict:** A safety rating (Low Risk, Medium Risk, High Risk, or Blocked) backed by real evidence.
- **Reversible Quarantine:** Backing up, turning off, and watching for 14 days before final deletion, with 1-click rollback.